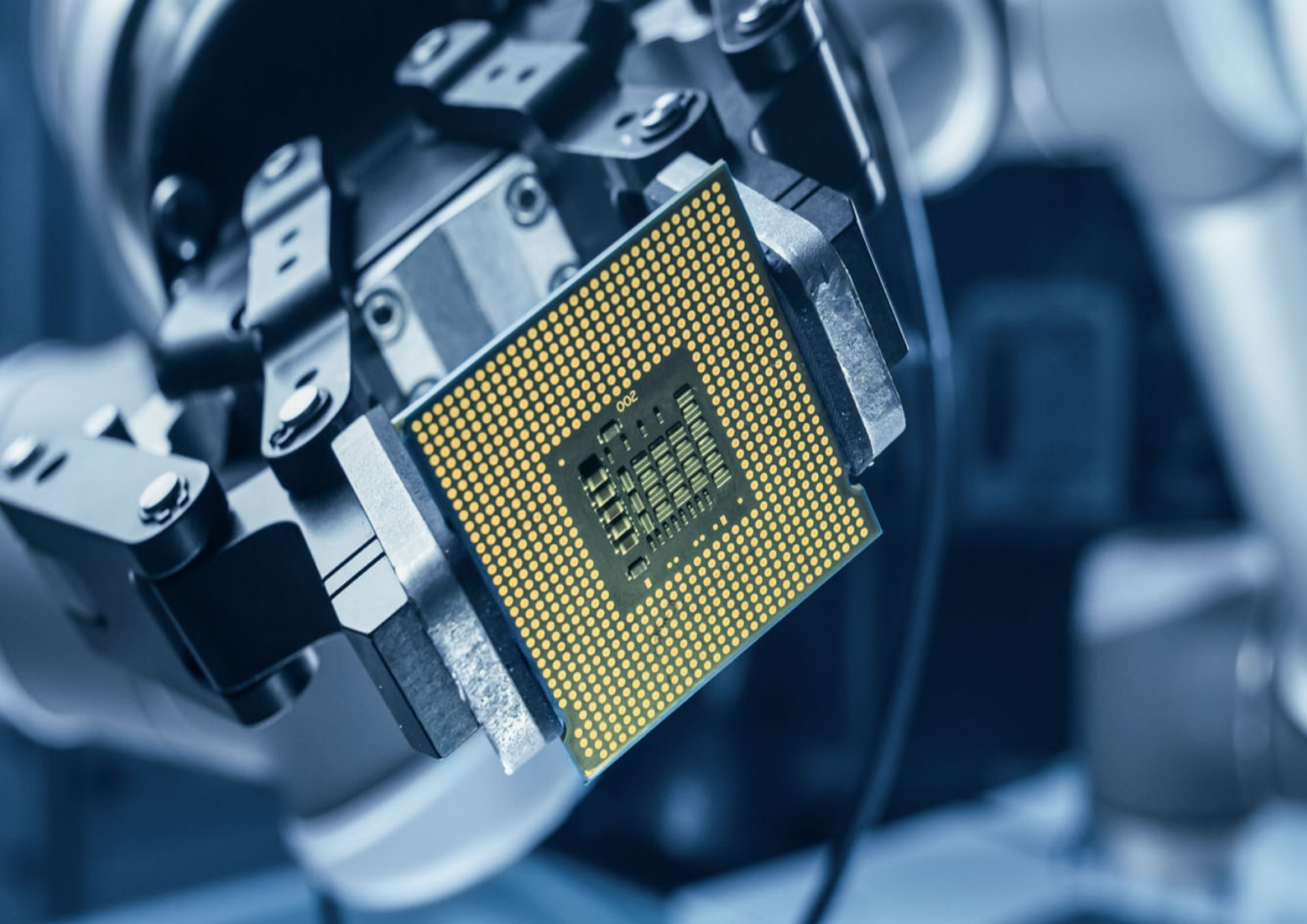




Análisis de amplificador cascode con BJTs

Agosto, 2023

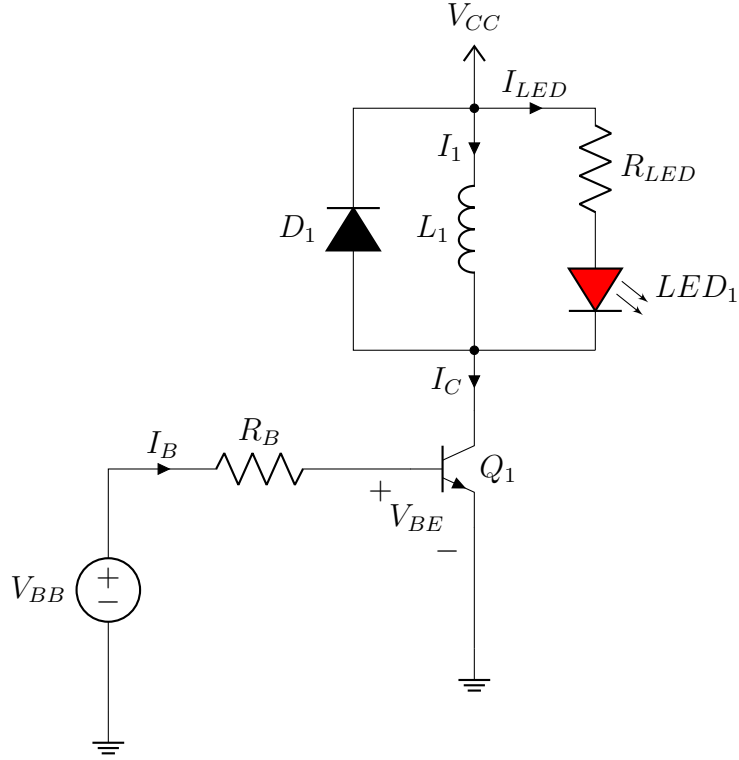


Figure 1: Transistor Switch

$$V_{BB} - R_B I_B - V_{BE} = 0$$

$$R_B = \frac{V_{BB} - V_{BE(sat)}}{I_B} \quad (1)$$

Where

$$I_B = \frac{I_C}{\beta_{forced}} \quad (2)$$

Assuming $\beta_{forced} = 20$, $V_{BB} = 3.3 \text{ V}$, $V_{CC} = 5 \text{ V}$, $V_{BE(sat)} = 0.7 \text{ V}$, $I_{LED} = 20 \text{ mA}$ and $I_1 = 170 \text{ mA}$ then:

$$I_B = \frac{I_C}{\beta_{forced}} = \frac{190 \text{ mA}}{20} = 9.5 \text{ mA} \quad (3)$$

Where

$$I_C = I_1 + I_{LED} \quad (4)$$

Then

$$R_B = \frac{3.3 \text{ V} - 0.7 \text{ V}}{9.5 \text{ mA}} = 273.7 \Omega \approx 270 \Omega \quad (5)$$

Finally

$$R_{LED} = \frac{V_{CC} - V_{LED} - V_{CE(sat)}}{I_{LED}} = \frac{5 \text{ V} - 2 \text{ V} - 0.2 \text{ V}}{20 \text{ mA}} = 140 \Omega \quad (6)$$